

Sprint 1 — Research framing and data foundation

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Due: October 7

The group must create a research foundation that is specific enough to implement and evaluate. The Sprint 1 review must include:

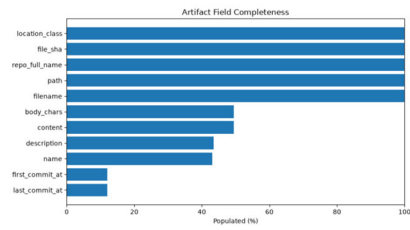
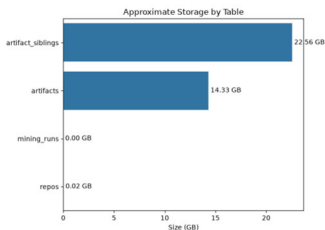
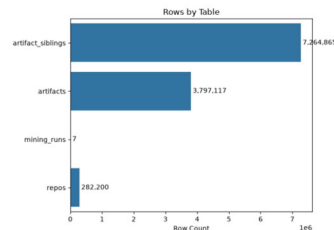
- A selected MSR-inspired topic and a precise research question.
- Motivation, expected contribution, and at least one competing or simpler explanation.
- A defined unit of analysis, population, sample, variables, and outcome measures.
- Dataset acquisition or sampling instructions and a data dictionary.
- A first working data-loading and extraction pipeline.
- At least one exploratory table or visualization generated by code.
- A repository with issues, milestones, branch protection or equivalent review practice, and a first pull request.
- A retrospective explaining what the group learned and what changed in the backlog.

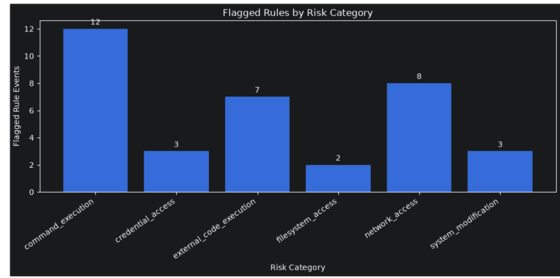
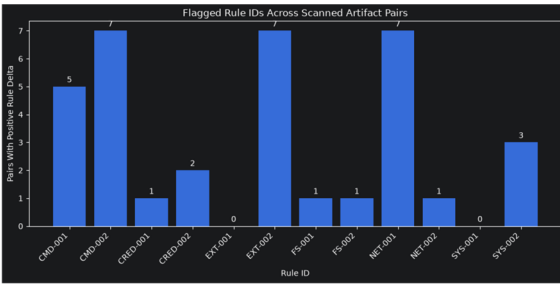
Sprint 1 acceptance criteria: The question is answerable with the selected data; the data path is documented; the pipeline runs on an approved sample; at least one result is generated; and the group has identified the primary risks and threats to validity.

1) Our team selected Research Question #4:

Research question: Do modified or reused skills introduce command execution, file-system access, network access, or other risky behavior that was absent from an earlier version or source artifact?

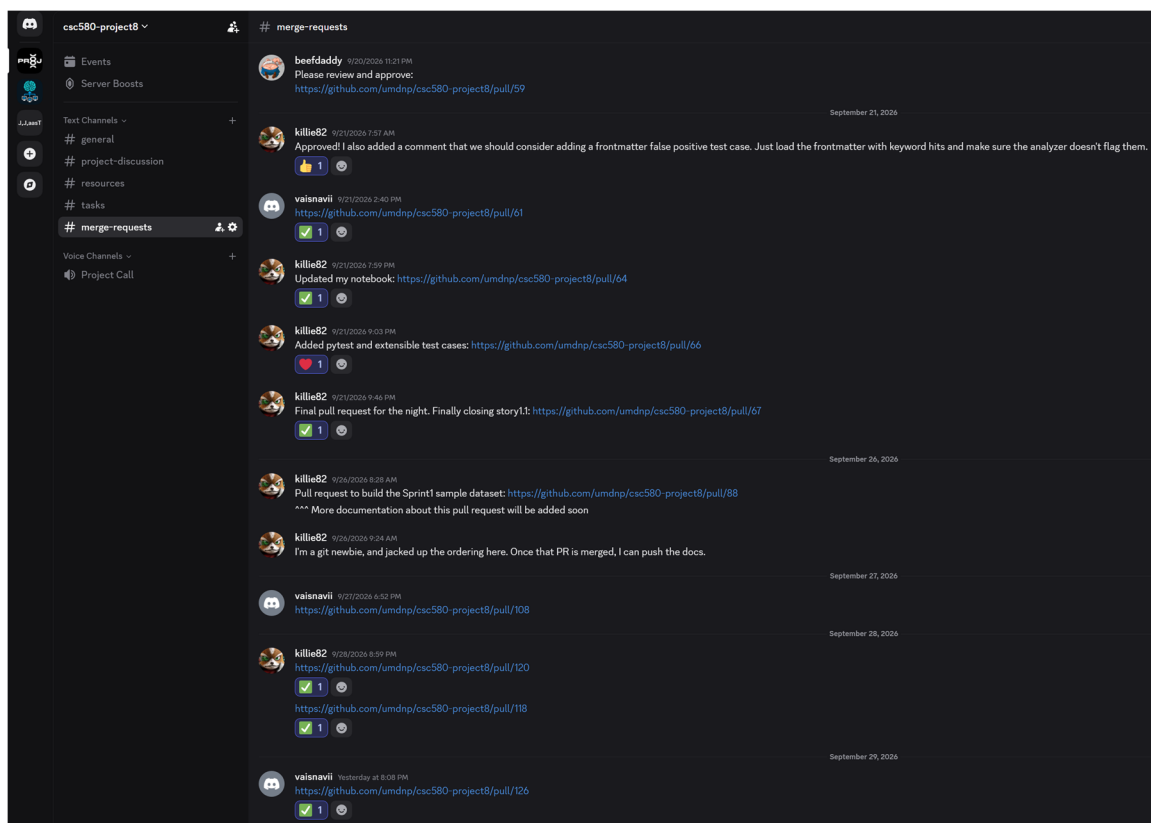
- 2) Our motivation, expected contribution, and competing explanation are documented in [RESEARCH_QUESTION.md](#).
- 3) Our unit of analysis, population, Sprint 1 sample, variables, and outcome measures are also defined in [RESEARCH_QUESTION.md](#).
- 4) Dataset acquisition, sampling instructions, and steps for building the analysis tables are documented in [data/README.md](#). The data dictionary is available in [DATA_DICTIONARY.md](#)
- 5) Our first working data-loading & extraction pipeline is documented in this [notebook](#). Instructions for running it are available in the accompanying [README.md](#)
- 6) Exploratory tables and visualizations were generated by code as part of our Sprint 1 analysis. Examples are shown below.





- 7) Our public GitHub repository is available at [csc580-project8](https://github.com/umdn/csc580-project8), and our Sprint 1 backlog is managed in the [Team 8 GitHub Project](#). The main branch is protected by rules that restrict deletions, require a pull request before merging, and block force pushes.

All pull requests are communicated on our team Discord server:



- 8) Our Sprint 1 retrospective, including what we learned and how it affected the backlog, is available in [docs/retrospective/sprint-1.md](#)
- 9) Our security threat model is documented in [THREAT_MODEL.md](#). The primary risks, limitations, and threats to validity identified during Sprint 1 are documented in [THREATS TO VALIDITY.md](#).